

● EASY

● GEOMETRY AND TRIGONOMETRY

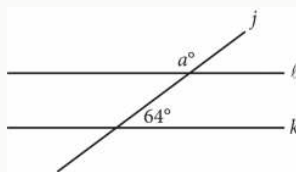
Lines, angles, and triangles

10 questions · ~15 min

03

Q1

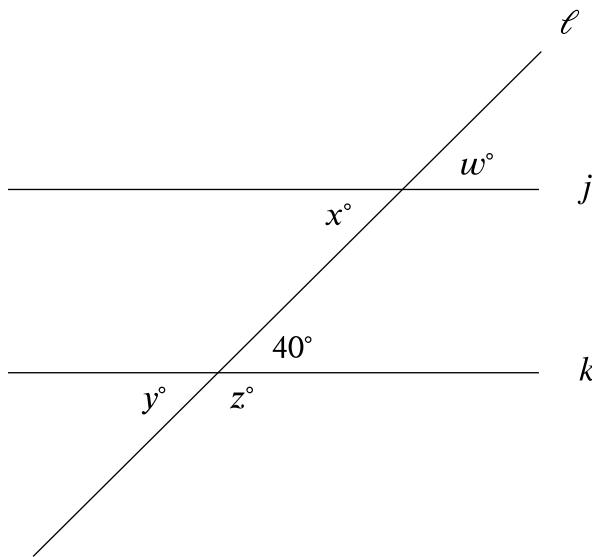
GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

- A. 26
- B. 64
- C. 116
- D. 154



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled script l , j , and k .
- Line script l intersects both line j and line k .
- At the intersection of line script l and line j , 2 angles are labeled clockwise from top left as follows:
 - Top right: w°
 - Bottom left: x°
- At the intersection of line script l and line k , 3 angles are labeled clockwise from top left as follows:
 - Top right: 40°
 - Bottom right: z°
 - Bottom left: y°
- A note indicates the figure is not drawn to scale.

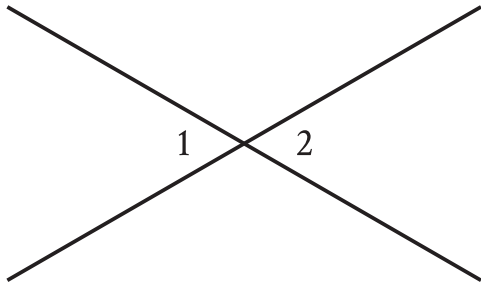
In the figure shown, line l intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

A. $w = 40$

B. $x = 140$

C. $y = 40$

D. $z = 140$

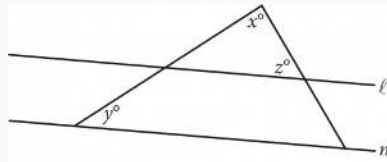


Note: Figure not drawn to scale.

- At the intersection of the 2 lines, the angles are labeled clockwise from left as follows:
 - Left: 1
 - Top: unlabeled
 - Right: 2
 - Bottom: unlabeled
- A note indicates the figure is not drawn to scale.

In the figure, two lines intersect at a point. Angle 1 and angle 2 are vertical angles. The measure of angle 1 is 72° . What is the measure of angle 2?

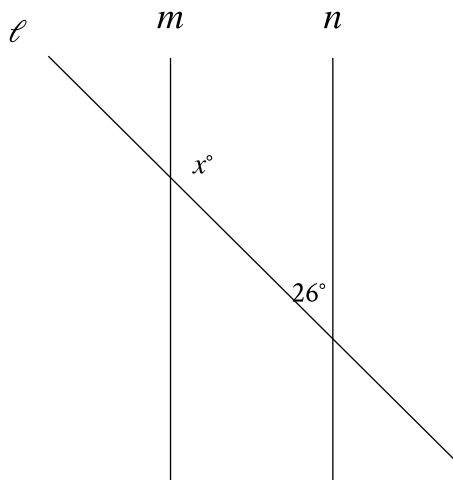
- A. 72°
- B. 108°
- C. 144°
- D. 288°



Note: Figure not drawn to scale.

In the figure above, lines l and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

- A. 120
- B. 100
- C. 90
- D. 80



Note: Figure not drawn to scale.

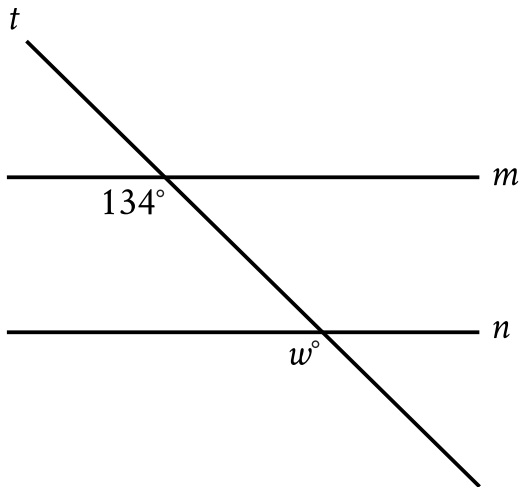
- Clockwise from top left, the 3 lines are labeled l , m , and n .
- Line l intersects both line m and line n .
- At the intersection of line l and line m , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line l and line n , 1 angle is labeled clockwise from top left as follows:
 - Top left: 26°
- A note indicates the figure is not drawn to scale.

In the figure shown, line m is parallel to line n . What is the value of x ?

- A. 13
- B. 26
- C. 52
- D. 154

In $\triangle RST$, the measure of $\angle R$ is 63° . Which of the following could be the measure, in degrees, of $\angle S$?

- A. 116
- B. 118
- C. 126
- D. 180



Note: Figure not drawn to scale.

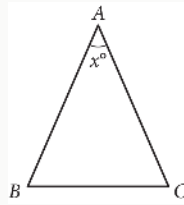
- Clockwise from top left, the 3 lines are labeled t , m , and n .
- Line t intersects both line m and line n .
- At the intersection of line t and line m , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: 134°
- At the intersection of line t and line n , 1 angle is labeled clockwise from top left as follows:
 - Bottom left: w°
- A note indicates the figure is not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134

Q8

GEOMETRY AND TRIGONOMETRY



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

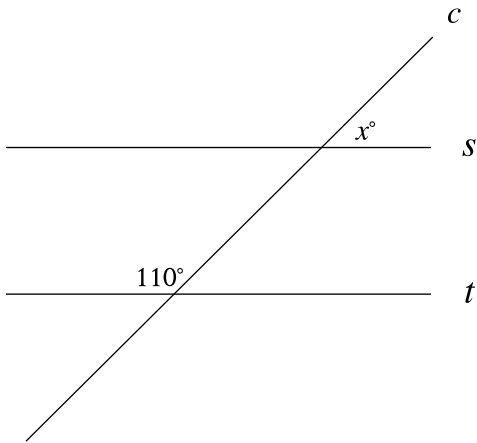
- A. 36
- B. 46
- C. 58
- D. 70

Q9

GEOMETRY AND TRIGONOMETRY

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled, c , s , and t .
- Line c intersects both line s and line t .
- At the intersection of line c and line s , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line c and line t , 1 angle is labeled clockwise from top left as follows:
 - Top left: 110°
- A note indicates the figure is not drawn to scale.

In the figure shown, line c intersects parallel lines s and t . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.